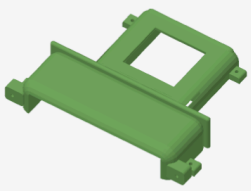
Creeper print kit & assembly

FIT PROTOTYPE · FRONT RED LED DISPLAY · REVERSIBLE SIDE CAMERA · NO HEARTS

Creeper: 8 parts to print

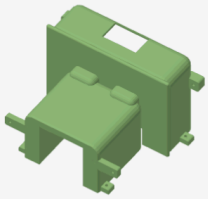
Front LED display · camera on either side · no hearts
Supplied print orientation shown. Individual views are not to the same scale.

01 **Body front**



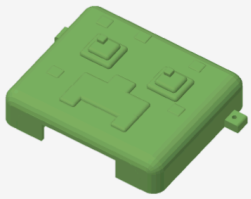
Print 1 · 196.0 × 155.0 × 36.97 mm

02 **Body back**



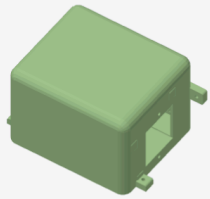
Print 1 · 196.0 × 155.0 × 121.97 mm

03 **Head front**



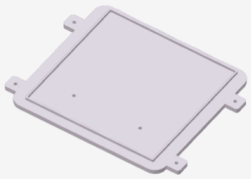
Print 1 · 154.0 × 111.88 × 37.99 mm

04 **Head back**



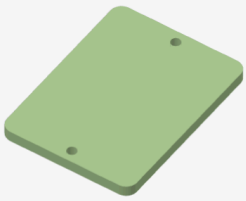
Print 1 · 160.0 × 111.88 × 90.88 mm

05 **Bottom tray**



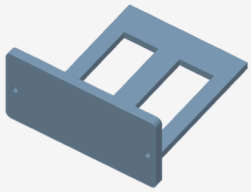
Print 1 · 196.0 × 153.0 × 8.0 mm

06 **Unused-port cover**



Print 1 · 48.0 × 64.0 × 4.0 mm

07 **Hub holder**



Print 1 · 84.0 × 88.0 × 42.0 mm

08 **Camera holder**



Print 1 · 35.0 × 41.0 × 62.0 mm

Fit prototype. Print one case section per plate. Use supports where Bambu Studio indicates them; actual printing and electronics fit are not yet verified.

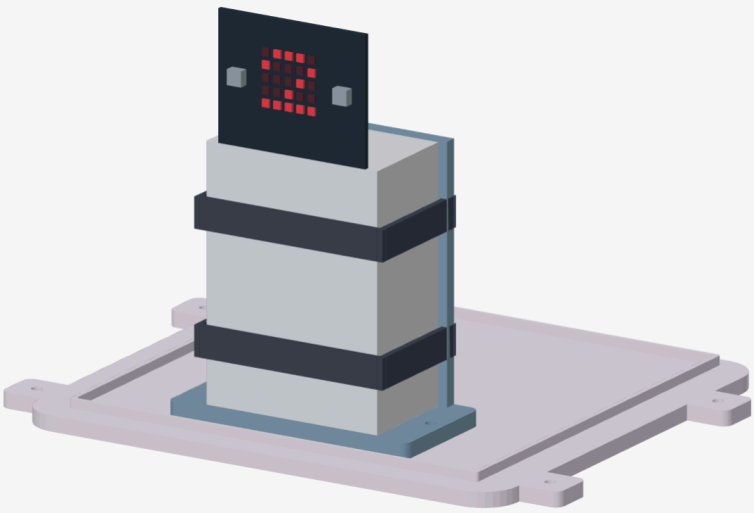
Hardware (not 3D printed)	Quantity
M3 × 20 mm machine screws	16
M3 × 12 mm machine screws	2
Standard M3 hex nuts (5.5 mm across flats)	18
10 mm-wide hook-and-loop straps, about 250 mm long	2
Small cable ties, no wider than 3.5 mm, about 150 mm long	2

Print the small hub-footprint test ring first. It checks only the published 59.8 × 82.4 mm footprint, not thickness or connectors. All 8 parts fit the P1S build envelope, but supports and real hardware fit have not been print-tested.

STLs are supplied in their print orientations. Use millimetres and 100% scale; use the matching P1S profile and inspect supports in Bambu Studio. Remove supports before fitting electronics. The original brick tower/base is replaced by this case.

1. Mount the hub and micro:bit

Parts 05 + 07 · 2 × M3×12 screws and nuts

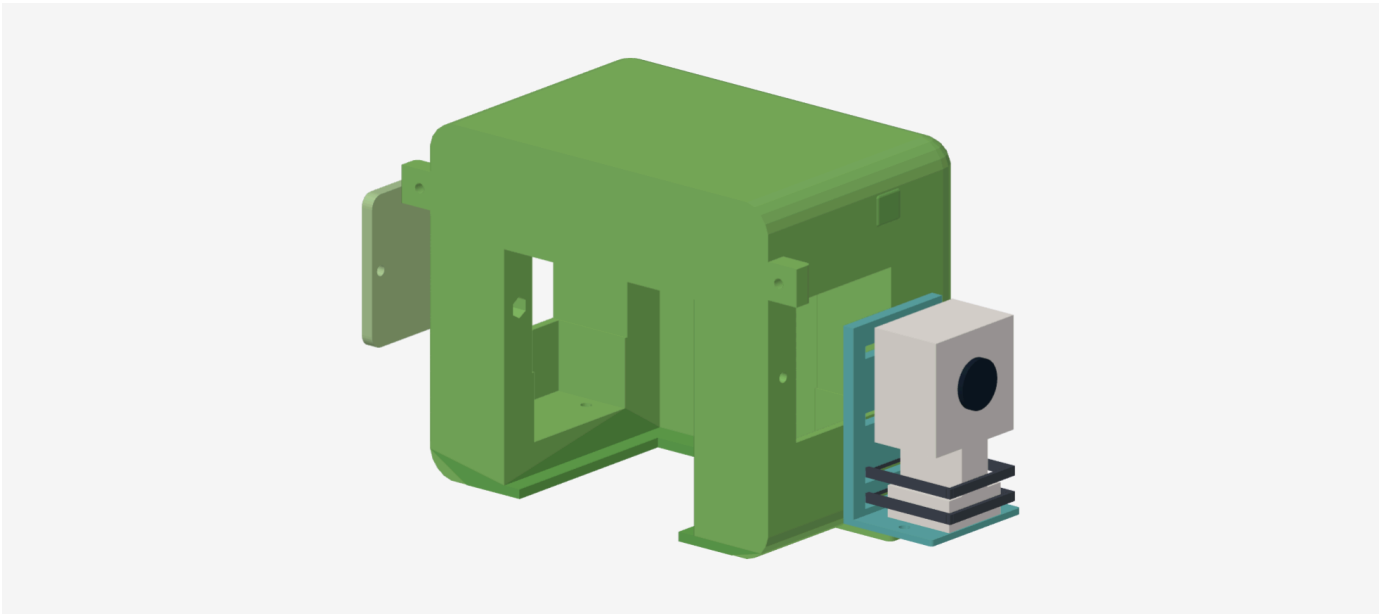


Bolt the blue-grey hub holder to the bottom tray. Keep the micro:bit plugged into the Nezha hub. Place the hub upright and use two 10 mm hook-and-loop straps through the tall slots. The red LEDs face forward. Move the straps clear of ports and the power button.

Power off/unplug before fitting. Check the real hub depth and connector position before strapping it in.

2. Choose the camera side

Parts 04 + 06 + 08 · 4 × M3×20 screws and nuts

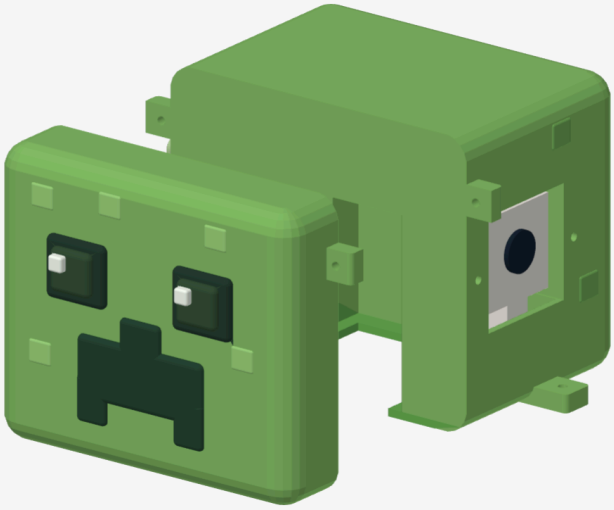


Right side is shown. Tie the camera mounting foot to the teal holder using two small cable ties through the lower slots. Bolt the holder into the chosen head pocket from underneath. For the left side, rotate the same holder 180° in plan. Bolt the green cover over the unused side; insert its nuts from inside the open head.

Do not force a fit or permanently glue the case. This layout still uses an assumed hub depth and camera thickness.

3. Close the head

Parts 03 + 04 · 2 × M3×20 screws and nuts

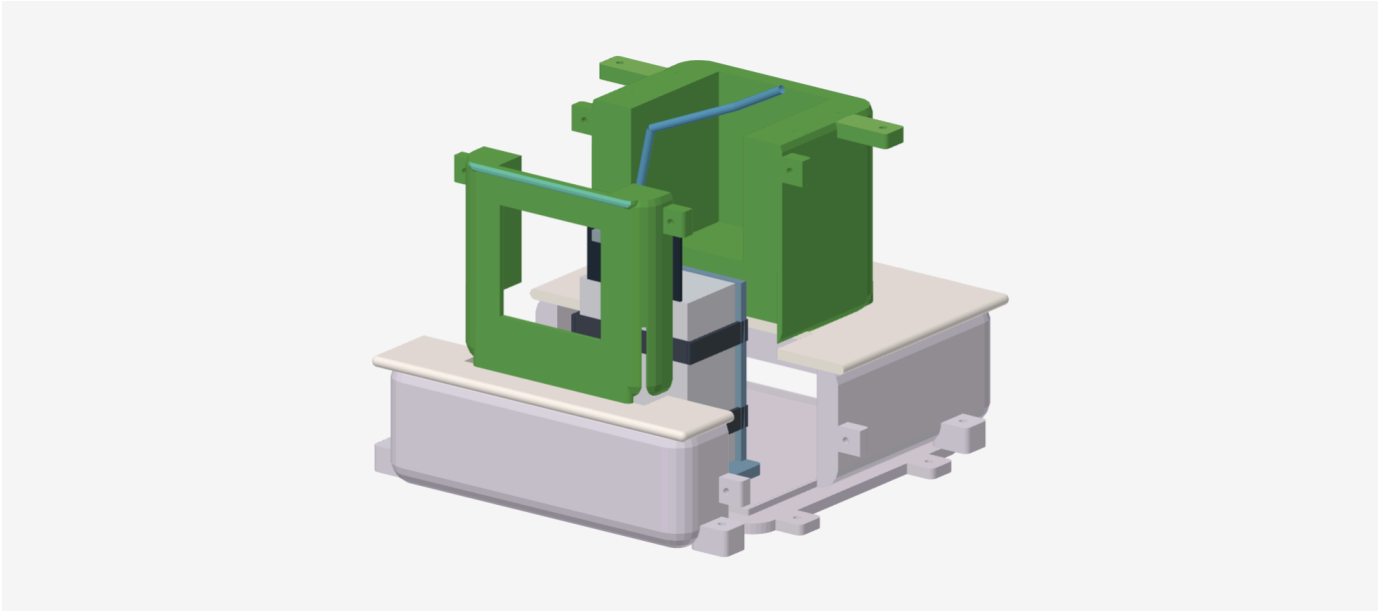


Connect the camera cable before closing. Keep the lens and screen clear of ties. Bring the head front and back together and fit the two screws through the side tabs above the camera ports. Keep the cable exiting through the open neck; do not trap it in the seam.

Do not force a fit or permanently glue the case. This layout still uses an assumed hub depth and camera thickness.

4. Close the body around the electronics

Parts 01 + 02 + 05 · 8 × M3×20 screws and nuts

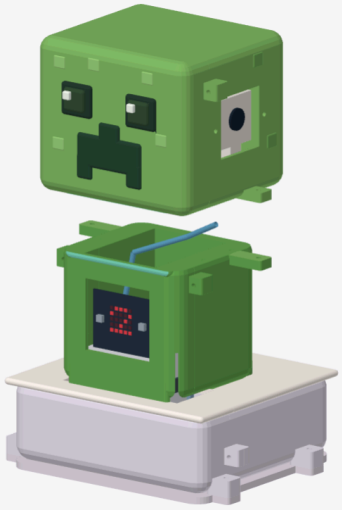


Route the existing compatible camera cable from an IIC port on the hub up through the neck, with slack. Bring the body halves around the hub/board assembly. Check the front viewing window shows all 25 LEDs. Fit four body-seam screws (two low, two at the shoulders), then four screws underneath the tray.

Do not force a fit or permanently glue the case. This layout still uses an assumed hub depth and camera thickness.

5. Join head and body

2 × M3×20 screws and nuts



Lower the head onto the body while guiding the cable into the neck. Align the two wide joint tabs toward the rear sides. Fasten each with a screw from below and a nut above. Do not force the joint if a wire or the recessed camera-holder screws prevent it seating.

Do not force a fit or permanently glue the case. This layout still uses an assumed hub depth and camera thickness.

6. Check before final tightening

Front LEDs visible · lens clear · no hearts



Power the electronics and check the whole LED display, A/B buttons, camera recognition and access to power/charging. Gently secure the joints only after the fit works. If a connector, strap or component clashes, stop and adjust the prototype before use. To swap camera side later, power off, lift and open the head, rotate/move the holder and move the blank cover.

Do not force a fit or permanently glue the case. This layout still uses an assumed hub depth and camera thickness.